MM-RFMNN: Multimodal Reliability-Aware Fuzzy Min-Max Neural Network

Comprehensive Algorithm Blueprint, Research Methodology & Implementation Plan

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Executive Overview

The Multimodal Reliability-Aware Fuzzy Min-Max Neural Network (MM-RFMNN) solves a fundamental failure mode in contemporary multimodal learning: catastrophic reliance on corrupted, uninformative, or missing modalities. While conventional deep early/late fusion models collapse when one channel is degraded, MM-RFMNN introduces:

1. Modality-Specific Hyperbox Quantization: Decoupled geometric representation spaces preserving individual modality semantics.

2. Leakage-Free Instance Reliability Estimation: Out-of-fold supervised meta-estimators scoring confidence, margin, and data quality.

3. Evidence-Calibrated Simplex Fusion: Temperature-calibrated arithmetic and logarithmic fusion with zero-imputation-free missingness masking.

4. Selective Classification & Abstention: Multi-criteria uncertainty bounds delivering certifiable risk-coverage guarantees.

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1. System Architecture & Information Flow

The MM-RFMNN pipeline processes K heterogeneous modalities through a decoupled, two-stage geometric and probabilistic network.

```
flowchart TD
    subgraph Inputs ["Input Space (Example i)"]
        X1["Modality 1: x_i^(1)"]
        X2["Modality 2: x_i^(2)"]
        XK["Modality K: x_i^(K)"]
        Q["Missingness Mask: q_i in {0,1}^K"]
    end

    subgraph Encoders ["Preprocessing & Encoders (Trained on Train Split)"]
        E1["f_1(x^(1)) -> z^(1) in [0,1]^d_1"]
        E2["f_2(x^(2)) -> z^(2) in [0,1]^d_2"]
        EK["f_K(x^(K)) -> z^(K) in [0,1]^d_K"]
    end

    subgraph FMNN_Layers ["Modality-Specific FMNN Hyperbox Banks"]
        H1["FMNN Bank H^(1): B_j^(1) = (V_j^(1), W_j^(1), c_j^(1))"]
        H2["FMNN Bank H^(2): B_j^(2) = (V_j^(2), W_j^(2), c_j^(2))"]
        HK["FMNN Bank H^(K): B_j^(K) = (V_j^(K), W_j^(K), c_j^(K))"]
    end

    subgraph Evidence_Gen ["Evidence & Signal Extraction"]
        S1["Class Evidence s^(1) in [0,1]^C & Margin / Membership"]
        S2["Class Evidence s^(2) in [0,1]^C & Margin / Membership"]
        SK["Class Evidence s^(K) in [0,1]^C & Margin / Membership"]
    end
```

```
    subgraph Reliability_Module ["Reliability Estimators (OOF Supervised)"]
        R1["g_1(u_i^(1)) -> r_i^(1)"]
        R2["g_2(u_i^(2)) -> r_i^(2)"]
        RK["g_K(u_i^(K)) -> r_i^(K)"]
        Gate["Presence Gating: r_i^(k) <- q_i^(k) * r_i^(k)"]
        Norm["Simplex Weight Normalization: alpha_i^(k) = r_i^(k) / sum(r_i)"]
    end

    subgraph Fusion_Layer ["Calibrated Evidence Fusion & Abstention"]
        Calib["Temperature Calibration: p_i^(k)(c) = Softmax(s_i^(k) / T_k)"]
        Fuser["Reliability-Weighted Fusion: p_i(c) = sum_k alpha_i^(k) * p_i^(k)(c)"]
        Uncertainty["Uncertainty Evaluation: Entropy, Margin, Disagreement, Total Rel
        Decision{"Abstention Criteria Met?"}
        Pred["Emit Prediction: y_hat_i = argmax p_i(c)"]
        Abstain["Abstain / Flag Low Confidence"]
    end

    X1 --> E1 --> H1 --> S1 --> R1 --> Gate
    X2 --> E2 --> H2 --> S2 --> R2 --> Gate
    XK --> EK --> HK --> SK --> RK --> Gate
    Q --> Gate --> Norm

    S1 --> Calib
    S2 --> Calib
    SK --> Calib
    Calib --> Fuser
    Norm --> Fuser

    Fuser --> Uncertainty --> Decision
    Decision -- No --> Pred
    Decision -- Yes --> Abstain
```

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2. Mathematical Formulation

2.1 Modality Encoding & Normalization

For K modalities, each input $x_i^{(k)} \in \mathbb{R}^{d_k}$ is preprocessed or encoded to $z_i^{(k)} = f_k(x_i^{(k)}) \in [0, 1]^{d_k}$.

To strictly prevent test-set leakage, min-max bounds or quantile scaling parameters are fitted exclusively on the training partition:

$$z_{ir}^{(k)} = \text{clip}\left(\frac{x_{ir}^{(k)} - \min_{j \in \mathcal{D}_{\text{train}}} x_{jr}^{(k)}}{\max_{j \in \mathcal{D}_{\text{train}}} x_{jr}^{(k)} - \min_{j \in \mathcal{D}_{\text{train}}} x_{jr}^{(k)} + \epsilon_{\text{scale}}}, 0, 1\right)$$

2.2 Hyperbox Representation & Fuzzy Membership

Each modality maintains a dedicated hyperbox bank:

$$\mathcal{H}^{(k)} = \{B_j^{(k)} = (V_j^{(k)}, W_j^{(k)}, c_j^{(k)}) \mid j = 1, \dots, M_k\}$$

where $V_j^{(k)} \in [0, 1]^{d_k}$ is the min-point, $W_j^{(k)} \in [0, 1]^{d_k}$ is the max-point, and $c_j^{(k)} \in \{1, \dots, C\}$ is the class label.

The dimension-wise outside distance of point z to hyperbox $B_j^{(k)}$ is:

$$d_{jr}^{(k)}(z) = \max(0, \, v_{jr}^{(k)} - z_r, \, z_r - w_{jr}^{(k)})$$

The total Manhattan outside distance is:

$$D_j^{(k)}(z) = \sum_{r=1}^{d_k} d_{jr}^{(k)}(z)$$

The continuous fuzzy membership function is:

$$\mu_j^{(k)}(z) = \exp\left(-\gamma_k \cdot D_j^{(k)}(z)\right), \quad \gamma_k > 0$$

where γ_k is the modality sensitivity parameter. Inside the box, $D_j^{(k)}(z) = 0 \implies \mu_j^{(k)}(z) = 1$. Outside the box, membership decays exponentially with Manhattan distance.

Class-level evidence for class $c \in \{1, \dots, C\}$ is the supremum across matching hyperboxes:

$$s_c^{(k)}(z) = \max_{j : c_j^{(k)} = c} \mu_j^{(k)}(z)$$

If no hyperbox exists for class c in modality k, $s_c^{(k)}(z) = 0$.

2.3 Feature-State Representation for Reliability Estimation

For example i and modality k, the meta-feature vector $u_i^{(k)}$ captures four orthogonal reliability facets:

1. Top Membership Strength:
   $$m_i^{(k)} = \max_{c \in \{1, \dots, C\}} s_c^{(k)}(z_i^{(k)}) \in [0, 1]$$
2. Class Margin (Separation Confidence):
   $$\Delta_i^{(k)} = s_{(1)}^{(k)}(z_i^{(k)}) - s_{(2)}^{(k)}(z_i^{(k)}) \in [0, 1]$$
   where $s_{(1)}^{(k)}$ and $s_{(2)}^{(k)}$ are the highest and second-highest class evidence scores.
3. Hyperbox Distance Sparsity:
   $$\delta_{\min, i}^{(k)} = \min_j D_j^{(k)}(z_i^{(k)})$$
4. Input Quality Indicators $Q_i^{(k)}$ (if present at test time, e.g., missing feature ratio, sensor variance, noise indicator):
   $$u_i^{(k)} = \left[m_i^{(k)}, \, \Delta_i^{(k)}, \, \exp(-\delta_{\min, i}^{(k)}), \, Q_i^{(k)}\right]^T \in \mathbb{R}^{d_u}$$

2.4 Reliability Meta-Estimator

Reliability $r_i^{(k)} \in (0, 1)$ is modeled via a parameterized function g_{ϕ_k}:

$$r_i^{(k)} = \sigma(g_{\phi_k}(u_i^{(k)})) = \frac{1}{1 + \exp\left(-(\mathbf{w}_k^T u_i^{(k)} + b_k)\right)}$$

(For non-linear relationships, g_{ϕ_k} is a 2-layer MLP with ELU activations and LayerNorm).

2.5 Leakage-Free Out-Of-Fold (OOF) Target Generation

A critical vulnerability of reliability modeling is training on in-sample predictions: because training error is artificially low, the reliability model becomes overconfident and fails to penalize subtle test corruptions.

To eliminate this leakage:

1. Partition training set $\mathcal{D}_{\text{train}}$ into V folds ($V=5$).
2. For each fold $v \in \{1, \dots, V\}$, train base FMNNs on $\mathcal{D}_{\text{train}} \setminus \mathcal{D}_v$.
3. Compute out-of-fold class predictions on $\mathcal{D}_v$:
   $$\hat{y}_{i, \text{OOF}}^{(k)} = \arg\max_c s_c^{(k), (-v)}(z_i^{(k)}), \quad i \in \mathcal{D}_v$$
4. Define the binary ground-truth correctness target:
   $$t_i^{(k)} = \mathbb{I}\left(\hat{y}_{i, \text{OOF}}^{(k)} == y_i\right)$$
5. Train g_{ϕ_k} on the unbiased dataset $\{(u_{i, \text{OOF}}^{(k)}, t_i^{(k)})\}_{i=1}^N$.

2.6 Dynamic Missingness Masking & Simplex Normalization

Let $q_i^{(k)} \in \{0, 1\}$ indicate presence (1) or absence (0) of modality k.

The masked reliability is:

$$\tilde{r}_i^{(k)} = q_i^{(k)} \cdot r_i^{(k)}$$

The simplex fusion weights $\boldsymbol{\alpha}_i = [\alpha_i^{(1)}, \dots, \alpha_i^{(K)}]^T$ are:

$$\alpha_i^{(k)} = \begin{cases}

\frac{\tilde{r}_i^{(k)}}{\sum_{l=1}^K \tilde{r}_i^{(l)}}, & \text{if } \sum_{l=1}^K \tilde{r}_i^{(l)} > \epsilon_{\text{rel}} \\

\frac{q_i^{(k)}}{\sum_{l=1}^K q_i^{(l)}}, & \text{if } \sum_{l=1}^K \tilde{r}_i^{(l)} \le \epsilon_{\text{rel}} \text{ and } \sum_{l=1}^K q_i^{(l)} > 0 \\

\frac{1}{K}, & \text{if all modalities are absent (total blackout)}

\end{cases}$$

2.7 Calibrated Evidence Fusion

To resolve modality scale differences, class scores are converted into calibrated probability distributions using temperature parameter $T_k > 0$ (tuned via Negative Log-Likelihood on validation data):

$$p_i^{(k)}(c) = \frac{\exp\left(s_c^{(k)}(z_i^{(k)}) / T_k\right)}{\sum_{q=1}^C \exp\left(s_q^{(k)}(z_i^{(k)}) / T_k\right)}$$

Primary Fusion: Reliability-Weighted Arithmetic Mixture

$$p_i^{\text{arith}}(c) = \sum_{k=1}^K \alpha_i^{(k)} p_i^{(k)}(c)$$

Property: Convex combination. Robust to single-modality catastrophic failure; if modality k becomes completely corrupt, $\alpha_i^{(k)} \to 0$ cancels its effect entirely.

Dual Fusion Alternative: Calibrated Log-Opinion Pool (Geometric Mixture)

$$\log \tilde{p}_i^{\text{geom}}(c) = \sum_{k=1}^K \alpha_i^{(k)} \log\left(p_i^{(k)}(c) + \epsilon_{\text{pool}}\right)$$

$$p_i^{\text{geom}}(c) = \frac{\exp\left(\log \tilde{p}_i^{\text{geom}}(c)\right)}{\sum_{q=1}^C \exp\left(\log \tilde{p}_i^{\text{geom}}(q)\right)}$$

Property: Unanimous veto capability. High penalty when any single reliable modality rejects class c.

2.8 Uncertainty Quantization & Multi-Tier Abstention

The decision rule computes prediction $\hat{y}_i = \arg\max_c p_i(c)$ alongside four distinct epistemic/aleatoric uncertainty indices:

1. Predictive Entropy:

   $$\mathcal{U}_{\text{ent}}(i) = -\sum_{c=1}^C p_i(c) \log\left(p_i(c) + \epsilon\right)$$

2. Confidence Deficiency:

   $$\mathcal{U}_{\text{conf}}(i) = 1 - \max_{c} p_i(c)$$

3. Modality Disagreement (Pairwise Jensen-Shannon Divergence):

   $$\mathcal{D}_{\text{JS}}(i) = \frac{1}{\binom{|\mathcal{K}_{\text{pres}}|}{2}} \sum_{j < k \in \mathcal{K}_{\text{pres}}} \left[\frac{1}{2} D_{\text{KL}}(p_i^{(j)} \parallel M_{jk}) + \frac{1}{2} D_{\text{KL}}(p_i^{(k)} \parallel M_{jk})\right]$$

   where $M_{jk} = \frac{1}{2}(p_i^{(j)} + p_i^{(k)})$.

4. Aggregate Available Reliability:

   $$\mathcal{R}_{\text{agg}}(i) = \sum_{k=1}^K \tilde{r}_i^{(k)}$$

Abstention Rule

The model abstains ($\hat{y}_i = \bot$) if:

$$\left(\mathcal{U}_{\text{conf}}(i) > \tau_{\text{conf}}\right) \lor \left(\mathcal{U}_{\text{ent}}(i) > \tau_{\text{ent}}\right) \lor \left(\mathcal{D}_{\text{JS}}(i) > \tau_{\text{disagree}}\right) \lor \left(\mathcal{R}_{\text{agg}}(i) < \tau_{\text{rel}}\right)$$

Thresholds $[\tau_{\text{conf}}, \tau_{\text{ent}}, \tau_{\text{disagree}}, \tau_{\text{rel}}]$ are optimized on validation data to meet user-specified risk-coverage budgets.

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3. Algorithm Specifications

Algorithm 1: Leakage-Free Two-Stage MM-RFMNN Training

```
Input: Multimodal dataset D_train = {(x_i^(1), ..., x_i^(K), y_i)}, validation set D_
       Expansion threshold theta_k for each modality k
       Modality dropout rate p_drop in [0, 0.5]
       Loss weights lambda_rel, lambda_modal

Output: Trained FMNN banks H^(k), calibrated scalers, reliability networks g_phi_k, t

Stage 1: Modality-Specific FMNN Expansion & Contraction
1: For each modality k in {1, ..., K}:
2:     Fit feature scaler f_k on D_train; compute z_i^(k) in [0,1]^d_k
3:     Initialize hyperbox set H^(k) = empty
4:     For each sample (z_i^(k), y_i) in D_train:
5:         Find existing box B_j^(k) of class y_i satisfying expansion limit theta_k:
               max(w_jr, z_ir) - min(v_jr, z_ir) <= theta_k for all r
6:         If box found:
7:             Expand V_j^(k) = min(V_j^(k), z_i^(k)), W_j^(k) = max(W_j^(k), z_i^(k)
8:         Else:
9:             Create new hyperbox B_new = (z_i^(k), z_i^(k), y_i); add to H^(k)
10:        Perform Simpson's overlap test & contraction against boxes of differing cl

Stage 2: Out-Of-Fold (OOF) Target Construction
11: Partition D_train into V folds {F_1, ..., F_V}
12: For fold v = 1 to V:
13:    Train temporary FMNN banks H_temp^(k) on D_train \ F_v
14:    For sample i in F_v and modality k:
15:        Compute membership s_c^(k, -v)(z_i^(k)) and prediction y_hat_i^(k, -v)
16:        Compute meta-feature u_i^(k)
17:        Set ground truth target t_i^(k) = 1 if y_hat_i^(k, -v) == y_i else 0
18: Collect unbiased meta-dataset D_meta^(k) = {(u_i^(k), t_i^(k))} for all i, k.

Stage 3: Reliability Model & Fusion Calibration
19: Train reliability estimators g_phi_k on D_meta^(k) minimizing Binary Cross-Entrop
        L_rel = - [t_i^(k) log(r_i^(k)) + (1 - t_i^(k)) log(1 - r_i^(k))]
20: Evaluate validation set D_val; fit temperature scaling parameters T_k via NLL:
        T_k* = argmin_T - sum_{i in D_val} log p_i^(k)(y_i; T)

Stage 4: End-to-End Fusion Fine-Tuning with Dynamic Dropout
21: For epoch = 1 to E_fusion:
22:    For mini-batch B in D_train:
23:        Sample random missingness mask q_i^(k) ~ Bernoulli(1 - p_drop)
24:        Inject synthetic corruption (Gaussian noise, masking) into one modality ra
```

```
        Compute calibrated p_i^(k), masked reliabilities r_tilde_i^(k), and weight
        Compute fused probabilities p_i(c)
        Compute composite objective:
            L_total = CrossEntropy(p_i, y_i) + lambda_rel * L_rel + lambda_modal *
        Update trainable fusion parameters via AdamW optimizer.

Stage 5: Abstention Boundary Calibration
On D_val, sweep grid over [tau_conf, tau_ent, tau_disagree, tau_rel]
Select thresholds maximizing retained accuracy at target coverage (e.g. >= 90%).
Return fully calibrated MM-RFMNN pipeline.
```

Algorithm 2: Dynamic Reliability-Aware Multimodal Inference

```
Input: Arbitrary test sample x = (x^(1), ..., x^(K)) with availability indicators q i
       Trained MM-RFMNN pipeline: {f_k, H^(k), g_phi_k, T_k, Tau}

If sum(q) == 0:
    Return Decision: ABSTAIN (Total modality blackout), Confidence: 0.0

For each modality k in {1, ..., K}:
    If q^(k) == 1:
        z^(k) = clip(f_k(x^(k)), 0, 1)
        For each box B_j in H^(k):
            D_j = sum_r max(0, v_jr - z_r^(k), z_r^(k) - w_jr)
            mu_j = exp(-gamma_k * D_j)
        For each class c in {1, ..., C}:
            s_c^(k) = max_{j: c_j = c} mu_j
        Extract meta-features u^(k) = [max_c s_c^(k), s_(1)^(k) - s_(2)^(k), exp(-
        Compute raw reliability r^(k) = sigmoid(g_phi_k(u^(k)))
        Compute calibrated distribution p^(k) = Softmax(s^(k) / T_k)
    Else:
        Set r^(k) = 0.0, p^(k) = Uniform(1/C)

Compute masked reliabilities r_tilde^(k) = q^(k) * r^(k)
If sum(r_tilde) < epsilon_rel:
    alpha^(k) = q^(k) / sum(q)  // Uninformative fallback
Else:
    alpha^(k) = r_tilde^(k) / sum(r_tilde)

Compute fused class distribution p(c) = sum_k alpha^(k) * p^(k)(c)
Compute uncertainty metrics:
        U_conf = 1 - max_c p(c)
        U_ent = -sum_c p(c) log(p(c) + 1e-12)
        D_JS = JensenShannonDivergence(available p^(k))
        R_total = sum(r_tilde)

If U_conf > tau_conf OR U_ent > tau_ent OR D_JS > tau_disagree OR R_total < tau_r
    Return Decision: ABSTAIN, Predicted_Class: argmax p(c), Risk_Flag: HIGH_UNCERT
Else:
```

```
26:    Return Decision: PREDICT, Predicted_Class: argmax p(c), Confidence: max p(c),
              Modality_Weights: alpha, Modality_Reliabilities: r_tilde,
              Winning_Hyperboxes: {k: argmax_j mu_j^(k)}
```

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4. Modular Software Architecture

The proposed Python/PyTorch codebase follows a strict object-oriented design isolating mathematical primitives, modality stores, and meta-learning modules.

```
mm_rfmnn/
|--- __init__.py
|--- base/
|   |--- hyperbox.py          # Vectorized HyperboxBank (V, W, class_labels)
|   |--- membership.py        # Simpson distance & continuous exponential membership
|   `--- contractions.py      # Vectorized overlap detection and dimensional contract
|--- encoders/
|   |--- base_encoder.py      # Abstract preprocessor interface
|   |--- tabular_scaler.py    # Training-fitted quantile/min-max normalizer
|   `--- pretrained_bridge.py # Fixed embedding extractors (ResNet, BERT, WavLM)
|--- reliability/
|   |--- meta_extractor.py    # Meta-feature extractor u_i^(k)
|   |--- oof_engine.py        # K-fold out-of-fold target cross-validation generator
|   `--- estimator.py         # PyTorch Module for g_phi_k (Linear/MLP)
|--- fusion/
|   |--- calibrator.py        # Temperature scaling optimizer via NLL
|   |--- fuser.py             # Arithmetic mixture & Log-opinion pool fusion layers
|   `--- abstention.py        # Risk-coverage curve and multi-threshold validator
|--- pipeline/
|   `--- mm_rfmnn_model.py    # Unified scikit-learn compatible estimator (fit, predi
`--- evaluation/
    |--- corruptions.py       # Deterministic noise suites (Gaussian, blur, dropout,
    `--- metrics.py           # ECE, Brier score, Macro-F1, AURC (Area Under Risk-Cov
```

Core Vectorized Hyperbox Class Specification

```
import torch
import torch.nn as nn
import numpy as np

class VectorizedHyperboxBank(nn.Module):
    """Vectorized, GPU-accelerated storage and distance-membership computation

    for FMNN hyperboxes within a single modality.
    """
    def __init__(self, in_features: int, gamma: float = 1.0):
        super().__init__()
        self.in_features = in_features
        self.gamma = gamma
```

```
        self.register_buffer("V", torch.empty((0, in_features), dtype=torch.float32))
        self.register_buffer("W", torch.empty((0, in_features), dtype=torch.float32))
        self.register_buffer("classes", torch.empty((0,), dtype=torch.long))

    def add_hyperbox(self, v: torch.Tensor, w: torch.Tensor, class_idx: int):
        self.V = torch.cat([self.V, v.unsqueeze(0)], dim=0)
        self.W = torch.cat([self.W, w.unsqueeze(0)], dim=0)
        self.classes = torch.cat([self.classes, torch.tensor([class_idx], device=self

    def forward_membership(self, Z: torch.Tensor) -> torch.Tensor:
        """Computes membership mu_j(z) for all input rows in Z against all hyperboxes

        Z: [BatchSize, in_features]
        Returns: [BatchSize, NumHyperboxes]
        """
        # Broadcast: [B, 1, D] vs [1, M, D]
        Z_exp = Z.unsqueeze(1)
        V_exp = self.V.unsqueeze(0)
        W_exp = self.W.unsqueeze(0)

        # Vectorized Manhattan outside distance
        d_lower = torch.clamp(V_exp - Z_exp, min=0.0)
        d_upper = torch.clamp(Z_exp - W_exp, min=0.0)
        D = torch.sum(d_lower + d_upper, dim=-1) # [B, M]

        return torch.exp(-self.gamma * D)

    def compute_class_evidence(self, Z: torch.Tensor, num_classes: int) -> torch.Tens
        """Returns class-level max membership: [BatchSize, num_classes]"""
        memberships = self.forward_membership(Z) # [B, M]
        batch_size = Z.shape[0]
        class_evidence = torch.zeros((batch_size, num_classes), device=Z.device)

        for c in range(num_classes):
            mask = (self.classes == c)
            if mask.any():
                class_evidence[:, c] = torch.max(memberships[:, mask], dim=1).values
        return class_evidence
```

---

5. Experimental Validation & Evaluation Suite

5.1 Benchmark Dataset Protocol

To rigorously evaluate across tabular, sensory, image, and text modalities, three diverse multi-source benchmarks are designated:

1. UCI Multimodal Sensor / Smartphone Activity (WISDM / PAMAP2):
   - Modality 1: 3-axis accelerometer.
   - Modality 2: 3-axis gyroscope.

- Modality 3: Physiological heart rate / tabular subject metadata.
- *Split protocol*: Subject-wise stratified split (Leave-Subjects-Out) to prevent identity leakage.

2. Multimodal Emotion / Sentiment (CMU-MOSEI / MELD subset):
- Modality 1: Pre-extracted audio acoustics (e.g. Wav2Vec2 / MFCC).
- Modality 2: Text transcription embeddings (e.g. RoBERTa frozen).
- Modality 3: Visual facial action units.
- *Split protocol*: Speaker/conversation-independent split.

3. Biomedical Multimodal Diagnostics (e.g., MIMIC-IV / Dermatology Image-Clinical):
- Modality 1: Clinical lab timeseries/tabular data.
- Modality 2: Clinical text summary or dermoscopic image features.
- *Split protocol*: Patient-stratified split.

5.2 Controlled Corruption Suites (Zero Test Leakage)

Models are subjected to 5 escalating severity levels ($l \in \{1, 2, 3, 4, 5\}$):

- Additive Gaussian Noise: $x_{\text{corrupt}}^{(k)} = x^{(k)} + \mathcal{N}(0, \sigma_l^2 \mathbf{I})$.
- Uniform Feature Masking / Drop: Randomly zeroing out fractions $\rho_l \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ of modality k.
- Adversarial / Extreme Drift: Constant sensor offset drift: $x_{\text{corrupt}}^{(k)} = x^{(k)} + \delta_l$.
- Complete Modality Dropout: Setting $q^{(k)} = 0$ for single and multi-modality combinations.

5.3 Comparative Baselines

1. Early Concatenation MLP: Features concatenated $\mathbf{z} = [z^{(1)}; \dots; z^{(K)}]$ into a deep classifier.

2. Uniform Late Fusion FMNN: Separate FMNN per modality fused via unweighted mean: $\alpha^{(k)} = \frac{1}{K}$.

3. Fixed-Weight FMNN: Modality weights set proportional to validation standalone accuracy: $\alpha^{(k)} \propto \text{Acc}_{\text{val}}^{(k)}$.

4. Early Concatenation FMNN: A single massive FMNN built over the concatenated space $\mathbb{R}^{\sum d_k}$.

5. Non-FMNN Reliability Baseline: Independent XGBoost/MLP classifiers per modality with learned softmax reliability fusion.

6. Proposed MM-RFMNN (Full Model): Vectorized hyperboxes + OOF-supervised reliability + temperature calibration + multi-metric abstention.

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6. Scientific Justification: Why This Plan is the Best

Design Dimension	Conventional Alternative	Proposed MM-RFMNN Strategy	Why Our Plan Dominates
Representation Space	Joint Concatenation ($z^{(1)} \oplus z^{(2)}$)	Decoupled Hyperbox Banks ($\mathcal{H}^{(k)}$)	Concatenation suffers from the curse of dimensionality and cross-scale distortion. Separate banks guarantee local geometric fidelity in each modality's natural basis.
Optimization Dynamics	Joint End-to-End Backprop through Boxes	Decoupled 2-Stage (FMNN Expansion + Differentiable Fusion)	Backpropagating min-max hyperboxes leads to vanishing gradients and box collapse. Simpson expansion is non-iterative, provably stable, and immune to local minima.
Reliability Supervision	In-Sample Training Correctness	Out-Of-Fold (OOF) K-Fold Target Generation	In-sample evaluation creates overfitted, overconfident reliability scores. OOF replicates true test generalization error, preventing catastrophic overconfidence.
Missing Modality Handling	Mean / Zero Imputation	Explicit Absence Gating ($q_i^{(k)} \cdot r_i^{(k)}$)	Imputing zeroes

creates artificial hyperbox hits and hallucinations. Our indicator mask zero-weights the entire channel, preserving mathematical integrity. |

| Evidence Alignment | Raw Membership Summation | Temperature-Scaled Softmax Normalization | Membership decays at different rates across different dimensions. Validation temperature scaling aligns all modalities into a consistent Bayesian probabilistic scale. |

| Selective Classification | Simple Softmax Thresholding | Multi-Tier Uncertainty (Entropy + Margin + Disagreement + Reliability) | Softmax probabilities are notoriously overconfident under out-of-distribution noise. Multi-tier checking catches modality conflict and low total evidence. |

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7. Systematic Risk Mitigation (Addressing the 10 Pitfalls)

```
classDiagram
    class ModalityMismatchMitigation {
        +TemperatureScaling T_k
        +IndependentQuantileTransforms
        +FixScaleDiscrepancy()
    }
    class LeakagePreventionMitigation {
        +KFold_OOF_Generator
        +PreprocessorsFitOnTrainOnly
        +ZeroTestContamination()
    }
    class MissingnessIntegrityMitigation {
        +BinaryPresenceMask q_k
        +SimplexRenormalization
        +NoZeroImputationHallucinations()
    }
    class DominantModalityMitigation {
        +AuxiliaryModalLoss lambda_modal
        +DynamicModalityDropout
        +PreventSingleModalityCollapse()
    }
    ModalityMismatchMitigation <|-- MM_RFMNN_Core
    LeakagePreventionMitigation <|-- MM_RFMNN_Core
    MissingnessIntegrityMitigation <|-- MM_RFMNN_Core
    DominantModalityMitigation <|-- MM_RFMNN_Core
```

1. Modality Scale Mismatch: Solved by strictly enforcing quantile normalization on inputs and tuning temperature scalars T_k via NLL on validation sets.

2. Reliability Leakage: Solved by implementing the V-fold Out-Of-Fold engine; the reliability estimator never observes hyperboxes fit on the evaluated sample.

3. Overconfidence in Outliers: Addressed by enforcing exponential distance falloff and dual epistemic entropy thresholds that trigger abstention if inputs fall far outside all hyperboxes.

4. Missing vs Noisy Distinction: Explicitly tracked by the binary mask $q^{(k)}$; absent modalities bypass distance computations entirely.

5. Dominant Modality Starvation: Mitigated by the auxiliary modality loss $\mathcal{L}_{\text{modal}}$ and dynamic modality dropout during Stage 4 training, forcing all modalities to maintain independent discriminative power.

6. Corruption Realism: Verified through held-out corruption types (e.g. trained on Gaussian noise, evaluated on sensor dead-band and quantization noise).

7. Patient/Subject Data Leakage: Handled via group-stratified splits (e.g., GroupKFold on Patient/Subject ID) across all experiments.

8. Class Imbalance: Hyperbox expansion thresholds θ_k and focal classification loss weights dynamically scale with inverse class frequencies.

9. Reliability vs Guarantee Claims: Formalized through empirical Risk-Coverage curves and Brier decomposition rather than ungrounded theoretical certainty claims.

10. Explainable Diagnostics: Every inference returns a structured attribution payload: winning hyperbox coordinates, distance margins, and normalized alpha weights, providing transparent geometric auditability.